

OSKALOOSA, IA, USA

WMO: 720351

Lat: 41.226N

Lon: 92.491W

Elev: 840

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.6	0.5	-11.2	3.1	-4.1	-6.7	4.0	1.3	32.0	25.1	28.6	23.5	8.8	320	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.5	92.0	75.2	90.0	74.6	87.7	73.0	78.9	88.4	77.1	85.9	75.2	84.0	9.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.5	142.8	85.8	73.6	128.8	82.5	72.5	124.3	81.1	43.1	88.8	41.0	85.8	39.2	84.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.7	24.0	20.5	-13.5	97.0	8.3	3.3	-19.5	99.3	-24.4	101.3	-29.1	103.1	-35.1	105.5		
			-12.8	81.7	7.0	1.4	-17.8	82.8	-21.8	83.6	-25.8	84.4	-30.8	85.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	24.0	24.0	39.9	51.4	63.1	72.5	75.0	72.3	66.2	53.2	40.0	29.4
	DBStd	20.62	12.45	11.80	11.75	9.96	8.70	5.50	5.98	5.33	7.87	9.53	10.50	11.21
	HDD50	3039	805	727	355	102	8	0	0	0	2	74	325	641
	HDD65	6093	1270	1147	781	419	142	8	4	6	81	382	749	1104
	CDD50	3429	0	0	42	143	413	675	775	692	486	175	26	2
	CDD65	1009	0	0	3	10	82	233	314	233	116	18	0	0
	CDH74	9609	0	0	25	177	790	2135	3168	2033	1087	192	2	0
	CDH80	3462	0	0	2	44	242	776	1302	682	369	45	0	0
Wind	WSAvg	9.3	11.3	11.0	11.2	12.0	10.6	8.4	6.0	5.1	6.7	8.7	10.5	10.1
Precipitation	PrecAvg	36.5	1.2	1.3	2.2	3.7	4.7	5.9	3.7	4.6	3.1	2.6	2.1	1.5
	PrecMax	52.8	2.5	3.6	4.7	8.3	11.0	12.4	11.1	11.3	8.9	6.5	5.9	3.3
	PrecMin	24.5	0.3	0.2	0.3	0.9	0.5	1.2	0.4	0.9	0.9	0.4	0.2	0.1
	PrecStd	7.9	0.6	0.8	1.3	1.5	2.3	2.8	2.8	2.9	1.8	1.7	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	59.5	78.1	86.0	91.2	92.9	98.1	93.4	91.9	86.1	72.9	60.0
		MCWB	45.3	49.7	63.0	64.7	71.2	74.0	76.0	74.8	73.5	69.0	58.1	53.3
	2%	DB	48.8	51.9	69.9	79.1	86.3	90.5	93.2	90.2	88.1	79.9	66.4	53.2
		MCWB	42.6	44.8	59.1	61.6	67.4	73.4	76.7	76.1	71.7	64.9	56.2	47.7
	5%	DB	43.7	44.5	63.5	73.3	81.9	88.1	90.2	87.3	84.4	74.7	60.8	48.5
		MCWB	39.3	39.5	55.4	58.0	65.5	71.8	76.4	74.3	70.4	62.6	51.6	44.8
	10%	DB	39.3	38.8	57.7	69.0	78.7	84.9	87.6	83.9	80.9	69.9	55.3	43.7
		MCWB	36.0	35.5	50.2	55.6	64.6	70.6	74.8	72.5	67.5	59.5	48.3	40.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.8	51.5	64.6	67.4	73.4	79.3	81.2	81.2	76.5	73.0	61.5	56.6
		MCDB	52.8	56.9	74.9	80.2	86.2	88.7	91.4	90.1	86.7	82.4	68.8	58.6
	2%	WB	43.1	45.0	60.8	64.0	71.1	76.6	79.3	78.5	74.3	68.7	57.4	48.9
		MCDB	48.3	50.0	67.3	74.5	82.4	85.7	89.7	86.4	84.4	75.0	64.1	51.2
	5%	WB	39.8	39.5	56.7	60.5	69.1	74.6	77.6	76.3	72.4	65.0	53.3	44.8
		MCDB	44.0	44.1	62.7	70.4	78.8	84.2	87.3	83.9	81.1	70.9	59.2	48.2
	10%	WB	36.3	35.8	50.7	57.9	67.0	72.8	75.7	74.1	70.1	61.3	49.3	40.1
		MCDB	39.4	38.7	57.3	67.0	75.1	82.1	84.4	81.7	77.0	68.7	55.3	43.7
Mean Daily Temperature Range		MDBR	16.6	16.6	18.2	21.8	20.5	19.9	20.5	20.5	22.8	21.4	19.0	15.6
		MCDBR	20.8	21.8	26.9	30.3	25.1	23.4	22.6	23.1	25.2	27.0	26.2	20.7
		MCWBR	15.6	15.7	16.3	16.1	11.0	10.5	10.1	11.5	10.8	13.6	16.5	15.9
	5% WB	MCDDBR	20.3	21.6	23.3	26.0	21.1	19.8	20.3	19.3	21.1	21.3	24.3	19.5
		MCWBR	15.4	16.0	15.0	15.9	10.8	10.7	10.4	11.2	10.4	13.8	17.2	16.3
Clear-Sky Solar Irradiance	taub	0.291	0.307	0.335	0.385	0.413	0.421	0.438	0.419	0.386	0.341	0.299	0.289	
	taud	2.419	2.372	2.388	2.262	2.254	2.280	2.218	2.290	2.332	2.458	2.503	2.441	
	Ebn at Noon	275	286	290	281	274	272	266	267	269	271	271	266	
	Edh at Noon	25	31	34	42	43	42	44	40	36	28	23	23	
All-Sky Solar Radiation	RadAvg	604	920	1199	1506	1765	1956	1969	1721	1471	980	663	506	
	RadStd	60	69	106	111	127	199	141	100	107	111	57	49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page